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A novel competitive swarm optimized RBF neural network model for short-term solar power generation forecasting

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ABSTRACT

Solar power is an important renewable energy resource and acts as a major contributor to replacing fossil fuel generators and reducing carbon emissions. However, the intermittent power output due to the uncertain solar irradiance significantly challenges the economic integrations of solar generation within the existing power system, which calls for effective forecasting methods to improve the solar prediction accuracy. In this paper, a novel improved radial basis function neural network model is proposed and applied in forecasting the short-term solar power generation. A recent proposed meta-heuristic approach named competitive swarm optimization is adopted to train the non-linear and linear parameters of the radial basis function neural network model. The proposed model has been validated in nonlinear benchmark functions and then employed in forecasting the solar power generation of a real-world case study in the Netherlands. Numerical results demonstrate that the proposed competitive swarm optimized radial basis function neural network model could obtain higher accuracy compared to other counterparts and thus provides a useful tool for solar power forecasting.

1. Introduction

The unprecedented global warming impact is deteriorating due to the increasingly heavy consumption of fossil fuel from both power generation and transportation sectors. The world has set up ambitious goals to control a temperature rise target within 1.5 $^{\circ}C$ agreed in the Paris Climate Conference 2015 [1]. To achieve this, substantial penetrations of renewable energy generations are required to replace the conventional fossil fuel based contributors and act as major power providers [2]. Solar power has long been one of the key renewable energy resources and is currently acting as the major power contributor in several developed countries. In the past few years, due to the continuous decrease of the cost, the world has seen a dramatical boost in the installation capacity of solar generation worldwide [3]. However, owing to the strong intermittent weather conditions, it is of significant difficulties to seamlessly integrate the solar power in various levels of power system. Accurate forecasting methods for solar power generation, therefore, are important for system operator in unit commitment and power dispatch and potential to bring remarkable improvement in terms of the economic and environmental performance.

Extensive research works have been proposed for solving various solar forecasting problems [4–6] including fundamental approaches, regression model, remote sensing model as well as artificial intelligence approaches. Under limited parameter data scenarios, artificial intelligence, in particular artificial neural network (ANN) based approaches have demonstrated powerful performance. Mellit and Pavan [7] utilized artificial neural network to forecast the power generation of a grid-connected PV plant at Trieste, Italy. Chen et al. [8] employed self-organized map method to classify the local weather type and adopted RBF neural network in the forecasting model establishment. Gala et al. [9] adopted hybrid machine learning methods including support vector regression, gradient boosted regression and random forest Regression to deliver 3-hour accumulated radiation forecasts. Pedro and Coimbra [10] presented a comparative study including various data-driven methods and concluded that the optimized based artificial neural network achieved the best performance. Deo et al. [11] developed wavelet-coupled support vector machine model to forecast global

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incident solar radiation. Yao et al. [12] proposed an echo state network integrated with principal component analysis in establishing the solar power forecasting model. In addition to the standard ANN, deep learning has also been adopted in forecasting solar power. Wen et al. [13] developed a deep recurrent neural network with long short-term memory units (DRNN-LSTM) for solar power and load forecasting. Srivastava and Lessmann [14] employed LSTM deep neural networks for irradiance forecasting in virtual PV sites. Though numerous neural network models have been proposed, it is required to make tremendous efforts in training the non-linear parameters of the neural network model.

Meta-heuristic algorithms (MAs) are powerful optimization tools that are immune from the problematic complexity and formulation. In the light of this, many featured MAs have been well adopted to optimize non-linear parameters of neural networks such as genetic algorithm (GA) [15], particle swarm optimization(PSO) [16,17], grey wolf optimizer [18], biogeography-based optimization (BBO) [19], monarch butterfly optimization (MBO) [20], whale optimization problem (WOA) [21], glowworm swarm optimization (GSO) [22], etc. Among various neural network models, radial basis function neural network is recognised for the simple structure and strong approximation ability for non-linear behaviours. Gao et al. [23] proposed a PSO based RBF classifier for two-class imbalanced problems, Yu and Duan [24] adopted artificial bee colony approach to train information granulation-based fuzzy radial basis function neural networks for image fusion applications. Wu et al. [25] adopted a hybrid PSO and GA method to optimize an evolving RBF neural networks for rainfall prediction Jafrasteh et al. [26] proposed a hybrid ABC-BP method to train the RBF neural network. Yang et al. [27] proposed a compact teaching-learning-based optimization to optimize feed-forward neural network (FNN) and RBF model. Rani et al. [28] utilized RBF model optimized by an enhanced PSO and differential search optimizer for wind speed prediction. Existing methods have achieved good results in approximating featured non-linear data. However, with the increasing number of hidden nodes, the parameters to be trained significantly boost, which challenges the state-of-the-art MAs.

The competitive swarm optimizer (CSO) is a recent proposed MA by Cheng and Jin [29]. Inspired by its ancestor PSO algorithm, the novel method updates the particles by substantially absorbing information from dual competitions. Competitive performance has been validated for the CSO algorithm particularly in solving large scale problems [30]. The canonical method and its variants have been successfully applied in solving various large-scale engineering problems [31,32]. In this paper, a novel CSO optimized RBF neural network is proposed and adopted in solving the solar generation forecasting problem. The key contribution of the paper has been listed below,

(1) A novel CSO optimized RBF model is adopted where the CSO method is for the first time adopted in training the non-linear parameters in RBF model.
(2) The proposed CSO based model training methods have been comprehensively compared with other state-of-the-art MA methods on featured non-linear modelling problems to validate the performance.
(3) A real-world case study of a Netherlands community PV generation is conducted for validating the model performance in solar forecasting, where monthly data models are established and investigated.

The rest of the paper is allocated as follows: the preliminary methodologies are presented in Section 2. The proposed CSO optimized RBF method is given in the Section 3, followed by the benchmark comparative study delivered in Section 4. The real-world case study of solar forecasting is presented in Section 5. Finally, Section 6 summarizes the article.

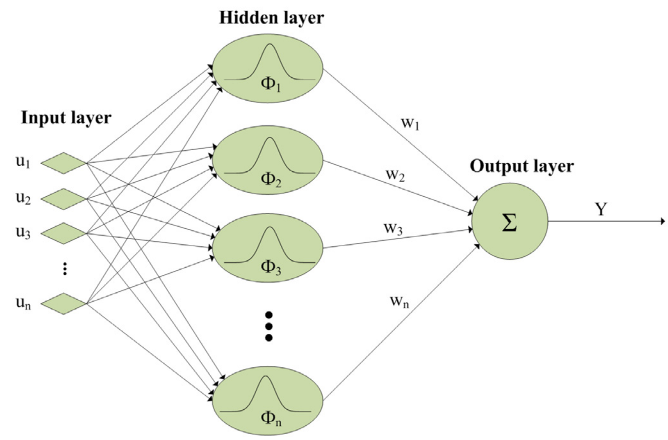

Fig. 1. RBF network structure.

2. Preliminaries

In this section, preliminary methods including the canonical radial basis function neural network and competitive swarm optimization is presented.

2.1. Radial basis function neural network

Conventional RBF neural network has been a multi-input and single-output (MISO) neural network structure, with the Gaussian functions severing as the activation functions. Three featured layers of RBF neural network, namely input, hidden and output layers are shown in Fig. 1.

$$y(t) = \sum_{i=1}^{n} w_i \cdot \phi_i(X) \tag{1}$$

The accumulated value $y(t)$ denotes the neural network model output at time lag t, whereas w_i corresponds to the linear output weight for the i^{th} node hidden layer. The radial basis function ϕ_i for input vector X is calculated as standard Gaussian function formulation as below,

$$\phi_i(X) = exp(-\frac{1}{2\sigma_i^2}\|X - c_i\|^2), i = 1, 2, ..., n \tag{2}$$

where σ_i and c_i denotes is the Gaussian distributed width and center of the i^{th} hidden node. The number of hidden layer is denoted as n. It could be noted that the featured non-linear function parameters denoted in the euclidean distance equations and denominator, which calls for effective optimization method to determine.

2.2. competitive swarm optimization

Competitive swarm optimization is one of featured bio-inspired algorithms [33] proposed by Cheng and Jin in 2015 [29]. The key learning operator mimics the bi-polar competition scheme where a couple of competitions are conducted between the winner and the loser, as well as between the mean value of the particle and the loser. The elegance of CSO method lies on that the competition scheme could fully activate all the variables information in each dimension, due to which it is particularly effective in solving large scale and complex optimization problems. The CSO algorithm and several variants have been adopted and proposed in various engineering applications [31,34–38]. The evolutionary logic of CSO method is shown as below,

$$V_{l,k}(t+1) = R_1(k,t)V_{l,k}(t) + R_2(k,t)(X_{w,k}(t) - X_{l,k}(t)) + \phi\ R_3(k,t)(X'_k(t) - X_{l,k}(t)). \quad (3)$$

$$X_{l,k}(t+1) = X_{l,k}(t) + V_{l,k}(t+1). \quad (4)$$

where $X_{w,\,k}(t)$ and $X_{l,k}(t)$ represent the position of the winner and the loser respectively. In each iteration, a new updated velocity denoted by $V_{l,k}(t+1)$ is composed of three parts: a randomly scaled velocity, a randomly scaled competition distance between the winner and loser of two random particles, as well as a weighted scaled competition distance between the loser and the mean value of the whole population $X'_k(t)$. The random scale are operated by random numbers $R_1(k,\,t)$, $R_2(k,\,t)$ and $R_3(k,\,t)$ all within (0,1). The weighting factor ϕ is the only parameter to be tuned in the algorithm, and it well can control the influence of $X'_k(t)$ in the optimization process. It should be noted that every particle is given one chance to take part in the competition. The winners of the competition will remain the same in the next generation, whereas the losers are updated by the updated velocity. In the algorithm, the tunable algorithm parameters have been limited with the only weighting factor one, which significantly reduces the tuning effort compared with other counterparts such as PSO and GA. Taking the advantage of the CSO method, the optimized RBF neural network is proposed in the next section.

3. Proposed CSO optimized RBF neural network

Similar to all the neural network model, during the applications of RBF neural network, both the model structure and model parameters are to be determined. Categorized by the determination flexibility, the model training procedures include three major types. Model Training Type 1 is to predetermine the model structure and non-linear parameters by trial and error approach, and then utilize least square or other approximation approaches to obtain the linear weights. Model Training Type 2 fixes the model structure such as input and hidden node number and allows optimization method to train the linear and non-linear parameters. Last but not least, Model Training Type 3 requires to simultaneously train the model structure and all the parameters. In this paper, Model Training Type 2 is adopted where CSO method is introduced in training the linear and non-linear parameters given a pre-determined neural network model structure.

In the training procedure of the RBF network, the root mean square error (RMSE) is adopted as the fitness function in the optimization, which is denoted as in (5) and (6)

$$min\ f = \sqrt{\frac{1}{N_m} \cdot \sum_{i=1}^{N_m} (\hat{y} - y_m)^2} \quad (5)$$

where $\hat{y}$ and y_m denote the prediction value and the measured data set respectively. Note that the formulation and all the parameters should be pre-set or determined before calculating the model output $\hat{y}$, which is denoted in equation

$$\hat{y}(t) = \sum_{i=1}^{n_h} w_i \cdot exp(-\frac{1}{2\sigma_i^2}\|X - c_i\|^2), i = 1, 2, ..., n. \quad (6)$$

In addition to the objective function, the training process of the parameters in RBF neural network could be regarded as an unconstrained optimization problem. Assuming that the numbers of input and hidden nodes are i and j, the decision variables could be encoded as below,

$$particle(i) = \begin{matrix}[c_{11}, c_{21}, ..., c_{j1}, \sigma_1, w_1, \\ c_{12}, c_{22}, ..., c_{j2}, \sigma_2, w_2, \\ ... \\ c_{1k}, c_{2k}, ..., c_{jk}, \sigma_k, w_k]\end{matrix} \quad (7)$$

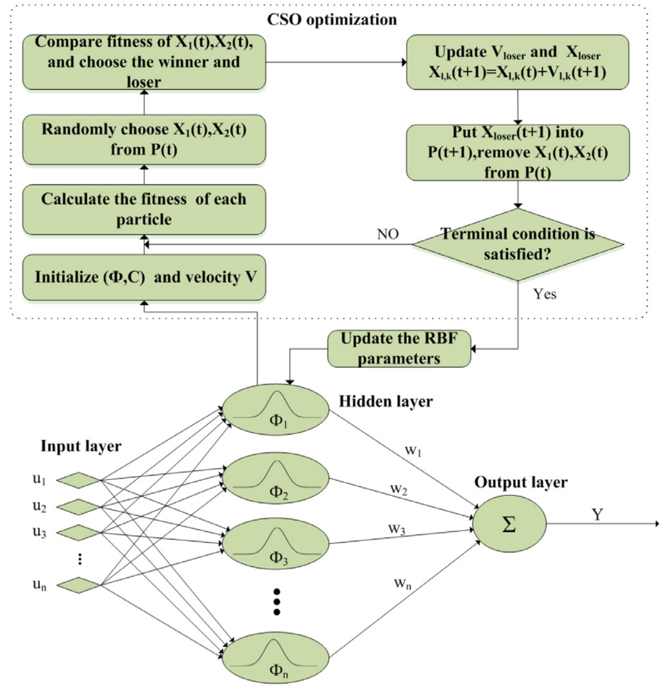

Fig. 2. Procedure of CSO optimized RBF network.

It could be observed that the non-linear parameters in each hidden node and corresponding linear weighting factors have been fully encoded. The detailed procedure of the proposed CSO optimized RBF modelling method could be found in 2 detailed as below,

1. Initialization:
   (a) Select input data vectors and adopt into the RBF neural network;
   (b) Determine the number of hidden nodes of RBF neural network;
   (c) Encode the decision variables according to the preset structure;
   (d) Initialize the values for the decision variables of a whole population;
2. Optimization process:
   (a) Calculate the mean value $X'_k(t)$ of the whole population;
   (b) Divide the whole population into two groups with equal number of particles;
   (c) Randomly select one particle from each group Calculate the fitness function (5) for both particles and determine the winner and loser;
   (d) Update the loser according to (3)-(4);
   (e) Put the updated loser and the winner into to the next generation population;
   (f) Repeat 2-c to 2-e until all the particles have been selected.
3. Finalization process:
   (a) Check whether the iteration has been reached the maximum number;
   (b) Jump to 2-a if the maximum iteration is not arrived;
   (c) Terminate the process and select the best performance particle.

4. Benchmark test

To investigate the performance of the proposed CSO optimized RBF neural network, two typical non-linear systems, a smooth

Table 1
RBF network training results of test system 1.

Hidden Node	Algorithm	Training RMSE	Training STD	Validation RMSE	Validation STD
3	wPSO-RBF	7.737e-03	3.737e-03	1.567e-02	8.311e-03
	DE-RBF	9.488e-03	6.388e-03	1.585e-02	1.343e-02
	TLBO-RBF	6.6391e-03	1.502e-03	1.405e-02	2.601e-03
	GWO-RBF	5.410e-03	2.622e-03	1.303e-02	1.595e-03
	MFO-RBF	5.289e-03	1.933e-03	1.304e-02	1.525e-03
	CSO-RBF	4.988e-03	9.862e-04	1.281e-02	9.884e-04
5	wPSO-RBF	9.108e-03	6.442e-03	1.647e-02	1.079e-02
	DE-RBF	1.186e-02	7.330e-03	1.646e-02	8.304e-03
	TLBO-RBF	7.848e-03	2.191e-03	1.669e-02	8.754e-03
	GWO-RBF	5.370e-03	2.414e-03	1.288e-02	3.049e-03
	MFO-RBF	5.626e-03	1.423e-03	1.313e-02	3.253e-03
	CSO-RBF	4.893e-03	1.365e-03	1.286e-02	1.100e-03
7	wPSO-RBF	1.088e-02	1.267e-02	1.584e-02	1.146e-02
	DE-RBF	1.352e-02	1.043e-02	1.896e-02	1.951e-02
	TLBO-RBF	8.889e-03	3.793e-03	1.674e-02	6.600e-03
	GWO-RBF	5.537e-03	1.897e-03	1.244e-02	3.892e-03
	MFO-RBF	5.521e-03	2.078e-03	1.397e-02	4.494e-03
	CSO-RBF	4.385e-03	1.439e-03	1.263e-02	1.037-03
9	wPSO-RBF	1.524e-02	2.293e-02	2.633e-02	4.477e-02
	DE-RBF	1.606e-02	1.468e-02	1.769e-02	1.526e-02
	TLBO-RBF	1.250e-02	7.309e-03	2.057e-02	1.047e-02
	GWO-RBF	5.328e-03	1.925e-03	1.357e-02	7.782e-03
	MFO-RBF	5.951e-03	3.395e-03	1.461e-02	5.336e-03
	CSO-RBF	4.606e-03	1.723e-03	1.224e-02	4.801e-03

Table 2
RBF network training results of test system 2.

Hidden Node	Algorithm	Training RMSE	Training STD	Validation RMSE	Validation STD
10	wPSO-RBF	2.974e-02	4.051e-03	5.041e-02	6.481e-03
	DE-RBF	2.970e-02	4.336e-03	5.128e-02	5.563e-04
	TLBO-RBF	2.635e-02	7.518e-03	4.546e-02	1.489e-02
	GWO-RBF	1.692e-02	3.281e-03	2.710e-02	7.585e-03
	MFO-RBF	1.693e-02	3.161e-03	3.095e-02	5.075e-03
	CSO-RBF	1.473e-02	2.518e-03	2.421e-02	2.486e-03
20	wPSO-RBF	3.439e-02	2.743e-03	5.548e-02	4.436e-03
	DE-RBF	3.994e-02	6.505e-03	6.250e-02	9.982-03
	TLBO-RBF	2.747e-02	1.047e-03	4.226e-02	8.095e-03
	GWO-RBF	1.390e-02	6.028e-03	2.117e-02	8.992e-03
	MFO-RBF	2.385e-02	5.305e-03	3.636e-02	2.316e-03
	CSO-RBF	1.117e-02	1.169e-03	1.769e-02	3.874e-03
30	wPSO-RBF	3.367e-02	6.566e-03	5.453e-02	1.028e-02
	DE-RBF	4.578e-02	2.281-03	7.690e-02	7.608e-03
	TLBO-RBF	3.392e-02	4.137e-03	5.621e-02	6.983e-03
	GWO-RBF	1.434e-02	4.558e-03	2.808e-02	7.501e-03
	MFO-RBF	2.562e-02	7.316e-03	4.428e-02	1.218e-02
	CSO-RBF	1.214e-02	1.913e-03	2.152e-02	3.308e-03

system and a highly non-linear system [27], are adopted as the benchmark in algorithms comparison. The CSO optimized RBF neural network is compared with RBF neural network optimized by several popular methods including inertial weighted PSO (wPSO) [39] and DE/rand/1 algorithm [40], as well as some other recent proposed approaches including grey wolf optimizer (GWO) [41] and moth flame optimization (MFO) [42], under the same optimization structure. All the algorithm parameter settings followed the original settings in the references, where the parameter *Phi* in CSO method is set as 0.2 as in [29]. For both test systems, 60% dataset is employed as the training data while 40% data are adopted for validation.

Training system 1 is a smooth non-linear system $f = sin(2x)e^{-x}$, of which the dataset from $(0, \pi)$ with 0.03 interval is selected as the model input. The number of particles is set as 30 for each method and the maximum iteration number is 100. Due to that the system is fairly smooth, less number of hidden nodes should be sufficient to approximate the non-linearity. The RBF structure employs 3-H-1 type, where three input items including $x(t)$, $x(t-1)$, and $y(t-1)$ have been adopted as the input and the number of hidden nodes is *H*. To investigate the impact of the number of hidden nodes on the model accuracy, *H* has been selected form 3 to 9, where the value of hidden nodes ranges from 0 to 5. In each test scenario, 10 independent runs are adopted to eliminate the randomness and the mean and standard deviation values of the errors are shown as in Table 1.

It could be observed in the Table 1 that the CSO optimized RBF obtained the lowest error in all the test scenarios. With the increase of hidden number, the training error reduces first and then increases. When the hidden nodes number is 7, the training error achieves the smallest value on 4.385e-03 and the validation error is 1.263e-02, both ranks the best among all the counterparts. However, due to that the non-linearity of Training System 1 is fairly smooth, a simple network structure could well describe the system. The advantage of CSO on solving large scale problems is not well demonstrated.

In addition to the case study of Training System 1, Training system 2 describes a highly non-linear system recursive system shown as below,

$$y(t) = 0.5y(t-1) + 0.8u(t-2) + u(t-1)^2 - 0.05y(t-2)^2 + 0.5 + \xi(t),\ \ \xi(\cdot) \sim N(0, 0.05), \tag{8}$$

In the Training system 2, *t*, *u* and *y* denotes time series, system input and output. The adopted system is a non-linear autoregressive exogenous model associated with a Gaussian system noise *N*(0, 0.05). In this training system, a total of 500 data are sampled among which 350 of them are used for model training and 150 data samples are used as model validation. The input *u* is uniformly distributed ranging between [−1, 1]. The RBF model structure is set as 5-H-1 and the input items are selected as $u(t)$, $u(t-1)$, $u(t-2)$, $y(t-2)$ and $y(t-1)$. Similarly, three test scenarios are compared with the hidden nodes selected as 10, 20 and 30 respectively. The test results are shown in Table 2.

It could be observed in Table 2 that the RBF neural network optimized by CSO again obtained the best results in all the three test scenarios and the 30 hidden nodes case see the lowest training and validation errors. With the increasing number of hidden nods, the CSO based method show more apparent advantage in comparing with other counterparts. The dimension of decision variables has achieve 150 for 30 test system scenario where CSO performs far better than other approaches. Both benchmark tests have demonstrated the competitive performance of the proposed CSO optimized RBF method. It would then be utilized in modelling the solar generation.

5. Solar forecasting case study

In this section, the competitive performance of the proposed CSO optimized RBF neural network is adopted in forecasting a real world solar generation scenario. The data is collected from a community in the Netherlands where the solar panels are installed on the buildings. The data recorded every 15 minutes ranging from 1st May to the 31st December 2018 are adopted in the modelling. Previous studies [7,10] have considered various weather and other factors of solar irradiance. Given that the sensor investments for other factors may not be realistic for community household users, in order to achieve an online real time forecasting task with, this study only adopts historical solar power data as the model input. Moreover, the majority of solar forecasting techniques may categorize the models according to the weather condition such as sunny, cloudy, rainy. It would significantly improve the model accuracy however rely heavily on the accuracy of weather forecasting.

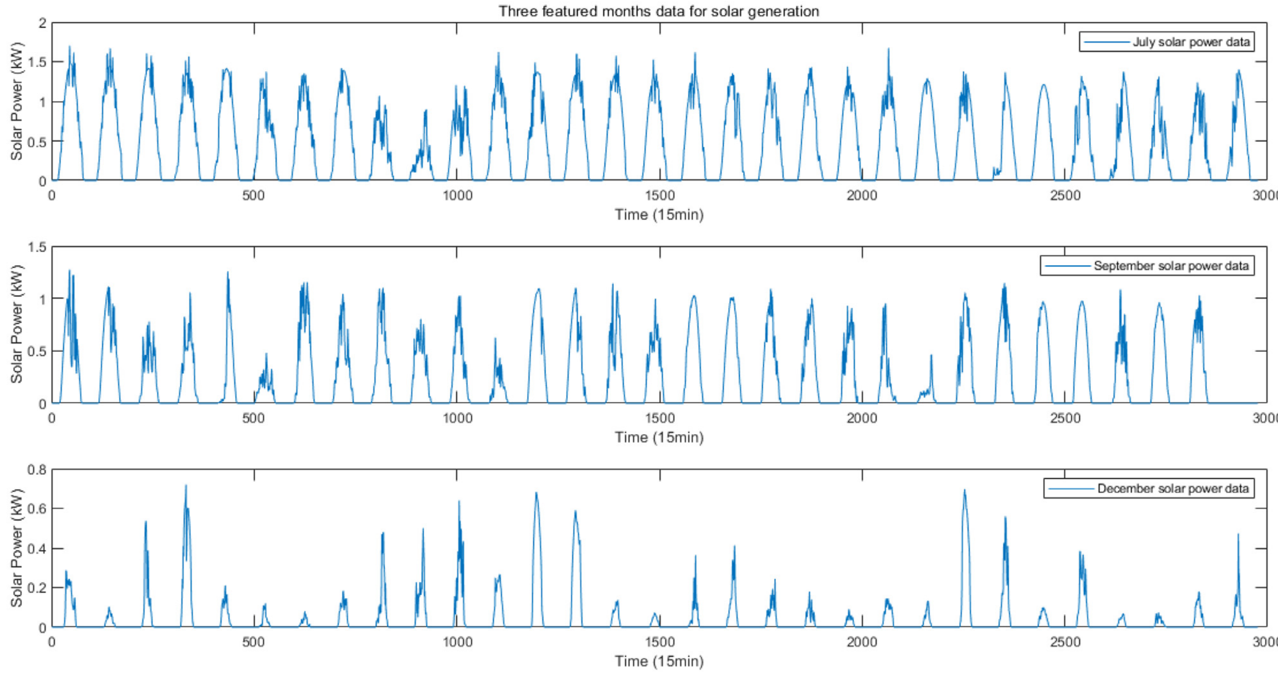

Fig. 3. Three featured months power data of solar generation.

In this paper, the model is trained by monthly data, which may consist of various featured weather conditions in the same month without losing generality. The test site is selected as a 39 houses community at Woerden, the Netherlands, where a whole year local weather and generation data has been fully recorded. Three featured months among the original dataset have been selected including a summer month-July, an autumn month-September and a winter month-December. The solar generation scenarios are denoted in Fig. 3.

It could be observed from Fig. 3 that in summer month July, the majority of the weather conditions are sunny with average large solar generations. In autumn month September, more cloudy weather appear and the uncertainty increases, which differs from the summer months. It could also be observed that in the winter month December, the solar generation has seen dramatic decrease due to the frequency rainy days, and nearly half of the days produced less than 0.2kW solar generation. To establish the corresponding models in the given featured month, the RBF models optimized by aforementioned heuristic counterparts including wPSO, DE, TLBO, MFO, GWO and CSO have been conducted. All the parameters of the algorithms are set as the same as in benchmark test, and the number of particles and function evaluations are set as 30 and 3000 respectively. 70% of the dataset are employed in training and 30% are adopted as validation. The model input adopted recursive terms $y(t-3)$, $y(t-2)$ and $y(t-1)$, and the number of hidden nodes are set as 10 according to preliminary trial and error study. Therefore, the employed RBF structure is 3-10-1, and again 10 independent runs are conducted for each test. The training results of the models for solar generation in three featured months are shown in Table. 3.

It could be observed in Table 3 that the CSO-RBF again achieves the best results in terms of both training and validation RMSE metric. The mean values of RMSE in summer, autumn and winter achieved by CSO-RBF are 3.843e-03, 4.131e-03 and 2.846e-03 respectively, successfully outperforms all the other counterparts. Comparing the three featured month models, there is no

Table 3
RBF network training results of solar generation in three featured months.

Hidden Node	Algorithm	Training RMSE	Training STD	Validation RMSE	Validation STD
July-Summer	wPSO-RBF	1.053e-02	4.863e-03	1.532e-02	6.6636e-03
	DE-RBF	1.153e-02	4.183e-03	1.734e-02	7.762e-03
	TLBO-RBF	5.017e-03	1.442e-03	8.394e-03	1.617e-03
	GWO-RBF	3.762e-03	2.428e-04	7.868e-03	5.687e-04
	MFO-RBF	3.988e-03	3.483e-04	7.773e-03	8.241e-04
	CSO-RBF	3.843e-03	5.742e-04	7.444e-03	1.149e-03
September-Autumn	wPSO-RBF	8.917e-03	3.340e-03	1.363e-02	5.839e-03
	DE-RBF	1.174e-02	2.433e-03	1.784e-02	2.863e-03
	TLBO-RBF	5.513e-03	6.215e-04	8.513e-03	1.590e-04
	GWO-RBF	4.112e-03	2.224e-04	8.535e-03	2.606e-04
	MFO-RBF	4.437e-03	8.727e-04	8.237e-03	6.878e-04
	CSO-RBF	4.131e-03	2.944e-04	8.149e-03	8.918e-04
December-Winter	wPSO-RBF	5.940e-03	4.713e-03	8.921e-03	6.443e-03
	DE-RBF	6.831e-03	4.536e-03	1.055e-02	7.526e-03
	TLBO-RBF	3.634e-03	4.550e-04	5.989e-03	6.455e-04
	GWO-RBF	2.956e-03	6.985e-04	5.915e-03	2.233e-04
	MFO-RBF	2.956e-03	2.538e-04	5.776e-03	5.319e-04
	CSO-RBF	2.846e-03	1.933e-04	5.693e-03	1.103e-03

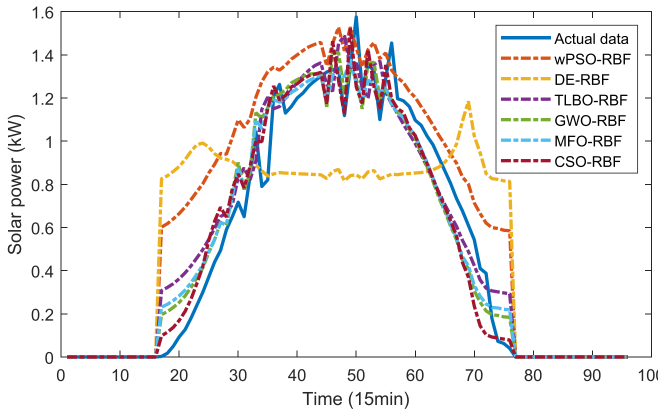

Fig. 4. Actual and prediction solar generation of a featured day in summer.

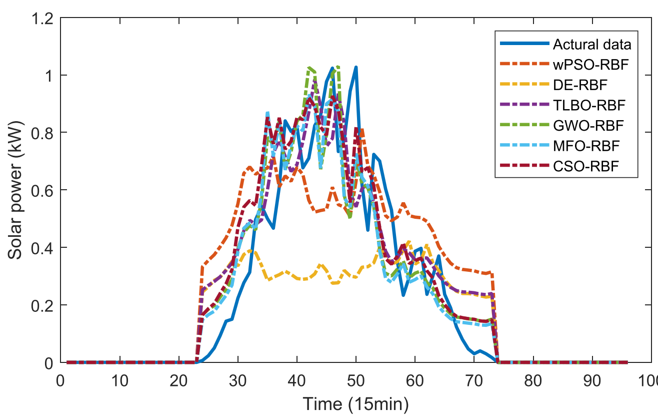

Fig. 5. Actual and prediction solar generation of a featured day in autumn.

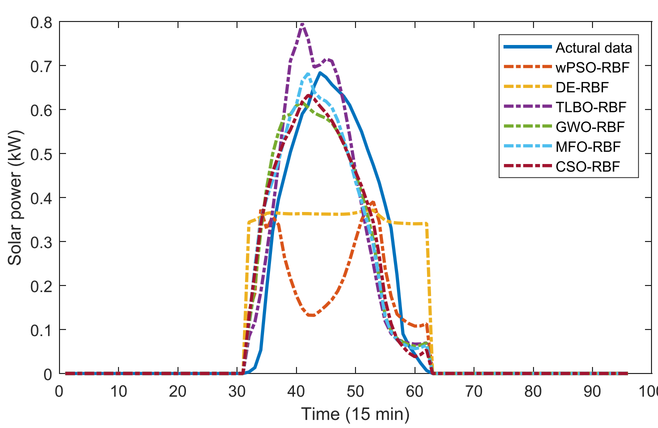

Fig. 6. Actual and prediction solar generation of a featured day in winter.

significant difference between the three scenarios, denoting that the proposed CSO-RBF method could achieve comparatively well model accuracy. To further investigate the model performance, typical-day forecasting scenarios of corresponding months have been demonstrated as in Fig. 4–6.

It could be clearly observed in 4–6 that the CSO-RBF achieved a comparatively accurate forecasting curve compared to other counterparts. Some methods such as DE-RBF performs terribly, possibly due to the lack of parameter training efforts. Different length of daylights have also been demonstrated in the figures and validates the model performance. As a result, real world numerical studies have proved that the proposed CSO optimized RBF method is an efficient modelling tool in forecasting solar generation. In addition, the proposed monthly based solar models could achieve satisfactory accuracy.

6. Conclusion and Future Work

In this paper, a novel competitive swarm optimization assisted radial basis function neural network is proposed for solar generation forecasting. Non-linear and linear parameters in RBF models could be simultaneously determined by the proposed model structure. The competitive performance of the proposed model has been validated in a couple of non-linear benchmark tests as well as a real world solar generation forecasting task. In addition, monthly solar models have been investigated and proved to be effective in reducing the necessity of more sensor investments.

Future work may be addressed on further improve the proposed CSO-RBF model by integrating model selection methods. In addition to the application on solar forecasting, the proposed CSO-RBF is potential to forecast the power demand, wind generation and all kinds of stochastic non-linear forecasting power and energy tasks, which would contribute in shaping a low carbon energy future.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work. There is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled “A novel competitive swarm optimized RBF neural network model for short-term solar power generation forecasting”.

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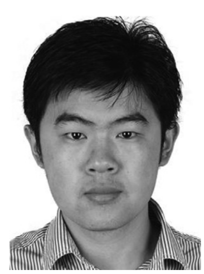

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